

EduWand Platform (DATABASE SCHEMA)

Prepared by: Fovea Infotech

1. Overview

This document defines the database schema for the two modules in this build phase: the Enrolment Growth Engine and the AI Module, along with the shared platform tables that support tenancy, users, and roles. Database engine is Amazon RDS for PostgreSQL. All tables include standard audit columns not repeated in every row below: id (UUID, primary key), created_at, updated_at, created_by, updated_by.

Tenancy is modelled Trust to School to Academic Year to Class to Section, per the PRD's multi-school requirement. Every table that holds school-specific data carries a school_id, and every school belongs to a trust_id, so data isolation between schools is enforced at the query layer.

2. Platform and Tenancy Tables

trust

Field	Type	Notes
name	text	Trust or society name
contact_email	text	
status	text	active, suspended

school

Field	Type	Notes
trust_id	uuid	Foreign key to trust
name	text	
board	text	CBSE, ICSE, State
address	text	
timezone	text	Default Asia / Kolkata
incumbent_erp	text	Name of existing ERP, used for CSV export mapping
status	text	onboarding, active, suspended

academic_year

Field	Type	Notes
school_id	uuid	Foreign key to school
label	text	For example 2026-2027

Field	Type	Notes
start_date	date	
end_date	date	
is_current	boolean	One current year per school

class_section

Field	Type	Notes
academic_year_id	uuid	Foreign key to academic_year
class_name	text	For example Grade 5
section_name	text	For example A

app_user

Field	Type	Notes
trust_id	uuid	Nullable, set if user is trust-level
school_id	uuid	Nullable if user is trust-level only
full_name	text	
email	text	Unique, used for login
phone	text	
role	text	front_desk, counsellor, teacher, admin, leadership
auth_provider_id	text	External identity provider reference
status	text	invited, active, disabled

A user may be scoped to a trust, one or more schools, or a single class, per the tenancy model. Where a user works across multiple schools within a trust, a separate user_school_access join table maps app_user to multiple school_id values.

3. Enrolment Growth Engine Tables

enquiry

Field	Type	Notes
school_id	uuid	Tenancy scope
contact_name	text	
contact_phone	text	
contact_email	text	Nullable

Field	Type	Notes
source	text	phone, walk_in, website, referral, event, social
grade_interest	text	Grade or board applied for
status	text	new, contacted, visit, application, admitted, enrolled, lost
lost_reason	text	Nullable, required if status is lost
owner_user_id	uuid	Foreign key to app_user, assigned counsellor
notes	text	
duplicate_of_enquiry_id	uuid	Nullable, set on merge
consent_captured	boolean	Messaging consent, DPDP

enquiry_stage_history

Field	Type	Notes
enquiry_id	uuid	Foreign key to enquiry
from_status	text	Nullable for the first entry
to_status	text	
changed_by_user_id	uuid	Foreign key to app_user
changed_at	timestamp	

follow_up_task

Field	Type	Notes
enquiry_id	uuid	Foreign key to enquiry
assigned_to_user_id	uuid	Foreign key to app_user
due_at	timestamp	
channel	text	sms, email
template_id	uuid	Foreign key to message_template
status	text	pending, sent, failed, cancelled
sent_at	timestamp	Nullable

message_template

Field	Type	Notes
school_id	uuid	Tenancy scope
channel	text	sms, email

Field	Type	Notes
name	text	
body	text	Template with placeholder variables
language	text	Default English

student_stub

Field	Type	Notes
school_id	uuid	Tenancy scope
source_enquiry_id	uuid	Foreign key to enquiry
full_name	text	
date_of_birth	date	
class_section_id	uuid	Foreign key to class_section
guardian_name	text	
guardian_contact	text	
admission_date	date	
fee_status	text	Placeholder only in this build phase

csv_export_log

Export uses a single fixed, standard column format across all schools. No per-school mapping table is needed, since a specific incumbent ERP is not tracked per school.

Field	Type	Notes
school_id	uuid	Tenancy scope
requested_by_user_id	uuid	Foreign key to app_user
schedule	text	Nullable, cron style if scheduled
run_at	timestamp	
row_count	integer	
status	text	success, failed
file_location	text	S3 path to the generated export file

4. AI Module Tables

lesson_plan

Field	Type	Notes
teacher_user_id	uuid	Foreign key to app_user
class_section_id	uuid	Foreign key to class_section
topic	text	
board	text	CBSE, ICSE, State
format	text	For example 5E
content	text	Generated lesson plan content
model_used	text	claude-sonnet
generated_at	timestamp	

research_report

Field	Type	Notes
teacher_user_id	uuid	Foreign key to app_user
topic	text	
content	text	Generated report content
model_used	text	claude-sonnet
generated_at	timestamp	

assignment

Field	Type	Notes
teacher_user_id	uuid	Foreign key to app_user
class_section_id	uuid	Foreign key to class_section
title	text	
personalisation_enabled	boolean	Teacher opt-in per assignment
status	text	draft, published, closed

assignment_personalisation_suggestion

Holds the system-suggested difficulty mix per student, pending teacher review. Never applied without an explicit approval recorded here, per the teacher-initiated fallback design.

Field	Type	Notes
assignment_id	uuid	Foreign key to assignment
student_stub_id	uuid	Foreign key to student_stub

Field	Type	Notes
suggested_difficulty_mix	jsonb	System suggestion, not yet applied
teacher_decision	text	pending, approved, overridden, opted_out
decided_by_user_id	uuid	Foreign key to app_user, nullable until decided
decided_at	timestamp	Nullable

submission

Field	Type	Notes
assignment_id	uuid	Foreign key to assignment
student_stub_id	uuid	Foreign key to student_stub
content	text	Submitted work
submitted_at	timestamp	

grade

Field	Type	Notes
submission_id	uuid	Foreign key to submission
ai_score	numeric	AI-assigned score
ai_feedback	text	Feedback and next steps
model_used	text	claude-haiku
teacher_override_score	numeric	Nullable
teacher_override_by_user_id	uuid	Nullable, foreign key to app_user
flagged_for_attention	boolean	AI-raised flag for a struggling student

ai_usage_log

Feeds the AI usage analytics view in the Admin Dashboard, and supports Bedrock cost tracking.

Field	Type	Notes
teacher_user_id	uuid	Foreign key to app_user
feature	text	lesson_plan, research_report, personalisation, grading
model_used	text	claude-sonnet, claude-haiku, nova-lite
input_tokens	integer	
output_tokens	integer	
created_at	timestamp	

5. Relationships Summary

- trust has many school
- school has many academic_year, enquiry, student_stub, app_user
- academic_year has many class_section
- enquiry has many enquiry_stage_history and follow_up_task, and may produce one student_stub
- assignment has many assignment_personalisation_suggestion and submission
- submission has one grade
- app_user, in the teacher role, generates lesson_plan, research_report, assignment, and produces ai_usage_log entries

6. Notes for Implementation

- All tables scoped to a school must filter by school_id in every query, enforced at the application layer, not left to client-side filtering
- assignment_personalisation_suggestion must never be read as an applied personalisation until teacher_decision is approved, this is the enforcement point for the DPDP-safe design
- ai_usage_log should be written asynchronously, from the worker instance, so AI calls are not slowed down by logging
- message_template.body should be treated as user-editable configuration, not hardcoded. The CSV export format itself is fixed and standard across all schools, not configurable per school